

Generalized Neyman-Pearson Lemma (GNP)

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Para Inf

Let $f_1, \dots, f_m, f_{m+1}: \mathbb{R}^n \rightarrow \mathbb{R}$ and integrable in $\mathbb{R}^n$ ($\int |f_i| d\mu < \infty$)

Suppose for some given constants $c_1, \dots, c_m$, there is a test ϕ ($\phi: \mathbb{R}^n \rightarrow [0,1]$) satisfying $\int \phi f_i d\mu = c_i$ for $i=1, \dots, m$

Consider the set $C_0 = \{ \phi: \int \phi f_i d\mu = c_i \forall i=1, \dots, m \}$

$$C = \{ \phi: \int \phi f_i d\mu \leq c_i \forall i=1, \dots, m \}$$

^{*}(i) $\exists$ a test $\phi_0 \in C_0$ that maximizes $\int \phi f_{m+1} d\mu$ over C_0

^{*}(ii) If $\exists$ constants $K_1, \dots, K_m$ s.t. the test

$$\phi^*(x) = \begin{cases} 1 & \text{if } f_{m+1}(x) > \sum_{i=1}^m K_i f_i(x) \\ 0 & < \end{cases} \quad \text{--- } (*)$$

satisfying $\int \phi^* f_i d\mu = c_i \forall i=1, \dots, m$ ($\phi^* \in C_0$)

then $\int \phi^* f_{m+1} d\mu \geq \int \phi f_{m+1} d\mu$ for any $\phi \in C_0$

(i.e. ϕ^* maximizes $\int \phi f_{m+1}$)

^{*}(iii) If $\exists$ constants $K_1, \dots, K_m \geq 0$ s.t. the test

$$\phi^*(x) = \begin{cases} 1 & \text{if } f_{m+1}(x) > \sum_{i=1}^m K_i f_i(x) \\ 0 & < \end{cases}$$

satisfying $\int \phi^* f_i d\mu = c_i \forall i=1, \dots, m$ ($\phi^* \in C_0$)

then $\int \phi^* f_{m+1} d\mu \geq \int \phi f_{m+1} d\mu$ for any $\phi \in C$

(iv) Consider the set $M = \{ (\int \phi f_1, \dots, \int \phi f_m): \phi: \mathbb{R}^n \rightarrow [0,1] \}$ is convex and closed. If $(c_1, \dots, c_m)$ is an interior point of M ,

then $\exists K_1, \dots, K_m$ s.t. ϕ^* in $(*)$ satisfy $\int \phi^* f_i = c_i \quad \forall i=1, \dots, m$.

If $\phi_0 \in C_0$ that maximizes $\int \phi f_{m+1}$ over C_0 then $\exists K_1, \dots, K_m$

$$\text{s.t. } \phi_0(x) = \begin{cases} 1 & \text{if } f_{m+1}(x) > \sum_{i=1}^m K_i f_i(x) \\ 0 & < \end{cases} \quad \mu\text{-a.e.}$$

Pf: (ii) Def $A = \{x: \phi^*(x) \neq \phi(x)\}$

$$\text{given } \begin{cases} \int \phi^* f_i = c_i \quad \forall i=1, \dots, m \\ \int \phi f_i = c_i \quad \forall i=1, \dots, m \end{cases}$$

$$\text{If } x \in A, \phi^* > \phi \Rightarrow f_{m+1}(x) \geq \sum_{i=1}^m K_i f_i(x) \quad (\text{b/c o/w } \phi^* = 0 \Rightarrow \phi > 0)$$

$$\text{If } x \in A, \phi^* < \phi \Rightarrow f_{m+1}(x) \leq \sum_{i=1}^m K_i f_i(x) \quad (\Rightarrow \phi^* < \phi)$$

$$g(x) = (\phi^*(x) - \phi(x)) \left(f_{m+1}(x) - \sum_{i=1}^m K_i f_i(x) \right) \geq 0 \quad \forall x \in A$$

$$\int g(x) = \int (\phi^* - \phi) (f_{m+1} - \sum K_i f_i) \geq 0$$

$$\Rightarrow \int \phi^* f_{m+1} - \int \phi f_{m+1} \geq \sum_{i=1}^m K_i \underbrace{\int (\phi^* - \phi) f_i}_{c_i - c_i} = 0$$

UMP Tests for Two-sided Hypotheses (in one-parameter exp family) (Application of GNP Lemma)

Thm: Suppose $X \sim f(x|\theta) = h(x)c(\theta)e^{\eta(\theta)T(x)}$; $\theta \in \mathbb{R}$ where $\eta(\theta)$ strictly increasing in θ

Want to test $H_0: \theta \leq \theta_1 \text{ or } \geq \theta_2$ vs. $H_1: \theta_1 < \theta < \theta_2$
(θ_1, θ_2 known)

(i) For testing $H_0: \theta \leq \theta_1 \text{ or } \geq \theta_2$ vs. $H_1: \theta_1 < \theta < \theta_2$

there exists a UMP size- α test given by

$$\phi^*(x) = \begin{cases} 1 & \text{if } c_1 < T(x) < c_2 \\ \gamma_i & \text{if } T(x) = c_i \text{ for } i=1,2 \\ 0 & \text{if } T(x) < c_1 \text{ or } T(x) > c_2 \end{cases}$$

where $c_1, c_2, \gamma_1, \gamma_2$ are determined from

$$E_{\theta_1} \phi^* = E_{\theta_2} \phi^* = \alpha$$

(ii) For any $\theta < \theta_1$ or $\theta > \theta_2$, ϕ^* minimizes $E_{\theta} \phi$ for any test ϕ satisfying $E_{\theta_1} \phi^* = E_{\theta_2} \phi^* = \alpha$

(iii) The power fn. $E_{\theta} \phi^*$ has a maximum at some $\theta_0 \in (\theta_1, \theta_2)$ and it decreases strictly as θ tends away from θ_0 in both direction unless $\exists t_1, t_2: P_{\theta}(T(x)=t_1) + P_{\theta}(T(x)=t_2) = 1 \forall \theta$

Pf: Sketch (i) $f_X(x|\theta) = h(x)c(\theta)e^{\eta(\theta)T(x)}$ then

$$f_T(t|\theta) = h_1(t)c(\theta)e^{\eta(\theta)t}$$

Consider $\phi(x) = \psi(T(x))$ for some fn ψ of T (which is sufficient)

Fix $\theta' \in (\theta_1, \theta_2)$ consider testing $H_0': \theta = \theta_1 \text{ or } \theta = \theta_2$ vs. $H_1': \theta = \theta'$

The problem: maximize $E_{\theta'}(\psi(T))$ s.t.

$$E_{\theta_1}(\psi(T)) = E_{\theta_2}(\psi(T)) = \alpha \quad \underbrace{\int \psi(t) f_T(t|\theta_1) dt}_{f_1} \quad \underbrace{\int \psi(t) f_T(t|\theta_2) dt}_{f_2} \quad \underbrace{\int \psi(t) f_T(t|\theta') dt}_{f_3}$$

By GNP Lemma $\phi^*(x) = \psi^*(T(x)) = \begin{cases} 1 & \text{if } f_T(t|\theta) > K_1 f_T(t|\theta_1) + K_2 f_T(t|\theta_2) \\ 0 & \end{cases}$

$$h_1(t) c(\theta') e^{\eta(\theta')t} > K_1 h_1(t) c(\theta_1) e^{\eta(\theta_1)t} + K_2 h_1(t) c(\theta_2) e^{\eta(\theta_2)t}$$

$$\Leftrightarrow \underbrace{K_1 \frac{c(\theta_1)}{c(\theta')}}_{a_1} e^{\underbrace{(\eta(\theta_1) - \eta(\theta'))}_{=b_1 < 0} t} + \underbrace{K_2 \frac{c(\theta_2)}{c(\theta')}}_{a_2} e^{\underbrace{(\eta(\theta_2) - \eta(\theta'))}_{=b_2 > 0} t} < 1$$

$$\Leftrightarrow a_1 e^{b_1 t} + a_2 e^{b_2 t} < 1$$

Case 1

$a_1, a_2 < 0$: not possible we will reject H_0 always (size not guaranteed)

Case 2

$a_1 < 0, a_2 > 0$ (vice versa) $g(t) = a_1 e^{b_1 t} + a_2 e^{b_2 t} \uparrow t$
 $\{g(t) < 1\} = \{t < c\}$

If $\phi^* = \mathbb{1}(t < c)$ monotone fn. in t

$E_{\theta}(\psi^*(T))$ monotone in θ
power fn.

In that case, $E_{\theta_1}(\psi^*(T)) = E_{\theta_2}(\psi^*(T))$ cannot be satisfied

$\therefore$ both $a_1 > 0, a_2 > 0$

$$g(t) = a_1 e^{b_1 t} + a_2 e^{b_2 t}$$

$$g'(t) = \underbrace{a_1 b_1}_{< 0} e^{b_1 t} + \underbrace{a_2 b_2}_{> 0} e^{b_2 t}$$

$$g''(t) > 0 \quad g \text{ strictly convex}$$

$$\Rightarrow \{g(t) < 1\} = \{c_1 < t < c_2\}$$

$$\phi^* = \begin{cases} 1 & \text{if } c_1 < T(x) < c_2 \\ 0 & \text{if } T(x) < c_1 \\ & \text{or } T(x) > c_2 \end{cases}$$

is MP size- α for $\theta = \theta_1$ or $\theta = \theta_2$ vs. $\theta = \theta'$

Note ϕ^* does not depend on $\theta' \Rightarrow \phi^*$ is UMP for $H_0': \theta = \theta_1$ or $\theta = \theta_2$
vs. $\theta \in (\theta_1, \theta_2)$

$$\text{T.S. } E_{\theta} \psi^*(T) \leq \alpha \quad \forall \theta < \theta_1, \theta > \theta_2$$

$$\text{Consider } \psi_0(T) = \alpha \quad E_{\theta_1}(\psi_0(T)) = \alpha = E_{\theta_2}(\psi_0(T))$$

$$\text{Using (ii) } E_{\theta}(\psi^*(T)) \leq E_{\theta}(\psi_0(T)) = \alpha \quad \forall \theta \in \Theta_0$$